

Connor Savugot

B.S. in Computer Engineering | North Carolina State University | Graduated May 2026

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TECHNICAL PROFILE

Low-level embedded firmware engineer specializing in real-time power, control, and communications software for microcontrollers and DSPs. Develops C/C++ firmware from peripheral bring-up and hardware debugging through firmware updates, system integration, and validation, with Embedded Linux as a supporting integration platform.

TECHNICAL SKILLS

Low-Level Firmware: Embedded C/C++, bare-metal and FreeRTOS development; peripheral bring-up, boot and update control

MCUs & DSPs: dsPIC33CK (16-bit DSC, Microchip HAL), PIC32 (32-bit MCU, FreeRTOS), TMS320 DSP (bare-metal C, TI HAL), STM32 (bare-metal C, STM32 HAL), MSP430FR2355 (bare-metal C)

Communications: CAN 2.0/CAN FD; J1939, DroneCAN; I2C, SPI, UART/RS-232; IPMI/IPMB

Systems, Debug & Test: TI AM62 Embedded Linux, Yocto/Arago, kernel and device trees, systemd, Docker; Python, Bash, Git/GitLab, JTAG, KiCad, SocketCAN/PCAN, oscilloscopes, logic/protocol analyzers, professional through-hole/SMT soldering

ENGINEERING EXPERIENCE

AEGIS Power Systems

Embedded Firmware Engineer

May 2026–Present

- Develop 16-bit dsPIC33CK firmware and bring up peripherals one subsystem at a time; isolate register, timing, configuration, and hardware faults with targeted tests, oscilloscopes, logic analyzers, schematics, and datasheets.
- Implemented and validated the VITA 46.11 Tier 2 management interface for a VPX power supply, providing IPMI 2.0 sensor, inventory, and event reporting over an I2C-based IPMB link to the chassis manager.
- Integrate battery hardware, CAN-connected devices, and Linux applications in embedded power systems; trace failures from power and wiring through bus traffic, services, configuration, and code.
- Support TI AM62 Embedded Linux integration: build Yocto/Arago images and kernels, update device trees, diagnose boot failures, and develop systemd services for external-watchdog recovery; use Docker for Windows/Linux tooling.

Embedded Systems Intern

May–Aug 2025

Computer Science Intern

May–Aug 2024

Selected work across both internships

- Developed firmware for a two-processor power system: a PIC32 running FreeRTOS managed I2C, SPI, CAN/J1939, and UART communications, while a TMS320 DSP ran bare-metal C firmware with TI hardware libraries for real-time power control.
- Developed DSP control firmware for transformer-coupled bidirectional buck-boost supplies using 12 linked PWM A/B pairs (24 outputs); configured each A channel's period/frequency, duty cycle, and relative phase while B followed or inverted A.
- Built host-side firmware-update tools for the same product: a C++ CLI accepted HEX files, while a Python TCP sender packetized updates with chunking and checksum validation; documented the update and toolchain workflows for engineering handoff.
- Implemented device-side update and recovery: the PIC32 staged firmware and a golden image in external SPI flash, programmed and verified the DSP over UART, and controlled the boot sequence.
- In 2024, created a restricted SSH/chroot interface with Bash and Python commands for secure remote inspection and control of TI Yocto/Arago hardware.
- Reverse-engineered RS-232 commands and checksums for poorly documented programmable resistive loads and relay controllers, then helped assemble the associated relay/load/power-supply test circuit.

TEAM Industries | Engineering Intern

Jun–Aug 2021 & Jun–Aug 2022

- Developed a repeatable Excel/VBA dimensional and spline-tolerance validation tool that automatically flagged out-of-spec parts; built a searchable catalog of 17,000+ tooling, part, and vendor records with operation, location, and revision traceability.

SENIOR DESIGN

EV Active Sensor Adapter | ST MCU, TI BQ ADC, CAN

NC State University

- Helped architect an EV sensor adapter integrating an STMicroelectronics MCU, TI BQ-family battery monitor/ADC, CAN transceivers, sensing, power, and telemetry on a custom PCB.
- Translated requirements and component trade studies into interface specifications and subsystem tests; coordinated design reviews, risk tracking, documentation, and bring-up across the team.

TEACHING & TECHNICAL LEADERSHIP

ECE 306 Teaching Assistant & Course Mentor | NC State University

Spring 2025–Spring 2026

- Supported three semesters: two in an unofficial role at about 10 hr/week and one as the official TA at 20 hr/week; developed study guides and delivered test preparation, labs, code/PCB debugging, project guidance, and ECE-program guidance.
- Taught students to build and debug MSP430FR2355-based vehicle subsystems spanning ADC sensing, motor control, GPIO/interrupts, Wi-Fi/serial links, LCDs, and custom PCBs; added supplemental lectures and question-driven support as the class average rose from C+ the prior semester to A-.

IEEE Open Project Space 2 | Lead Instructor

Fall 2025 & Spring 2026

- Led two semesters of a project-based electronics course from idea through demonstration; taught firmware, PCB/BOM workflows, and through-hole/SMT assembly while coaching students to solve their own design problems.

IEEE Soldering Workshops | Coordinator, Lecturer & Lab Instructor

- Led standalone workshops using custom through-hole and surface-mount flashing-light PCB keepsakes; taught professional soldering technique, safety, inspection, debugging, and rework.